

t-Test: Two-Sample Assuming Equal Variances

Statistical Methodology & Analysis of Historical Datasets (1969 vs. 1970)

1. Introduction & Theoretical Framework

The **Two-Sample t-Test Assuming Equal Variances** (also known as an *independent samples t-test* or *pooled-variance t-test*) is a parametric statistical method used to determine whether the means of two independent groups are significantly different from each other. This test assumes that the continuous dependent variable is normally distributed within each group and that both groups possess equal population variances ($\sigma_1^2 = \sigma_2^2$).

Hypotheses

The test evaluates a null hypothesis against an alternative hypothesis:

- **Null Hypothesis (H_0):** $\mu_1 - \mu_2 = d_0$ (typically $d_0 = 0$, indicating that the two population means are identical).
- **Alternative Hypothesis (H_1 or H_a):**
 - $\mu_1 \neq \mu_2$ (Two-tailed: means are simply different)
 - $\mu_1 > \mu_2$ or $\mu_1 < \mu_2$ (One-tailed: specify a directional difference)

Mathematical Formulation

When the variances of the two populations are assumed to be equal, a combined or **Pooled Variance** (s_p^2) is calculated to offer a single, more robust estimate of the common variance:

$$s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$$

Where s_1^2 and s_2^2 are the sample variances, and n_1 and n_2 are the respective sample sizes. Using this pooled variance, the test statistic t is computed as follows:

$$t = (\bar{X}_1 - \bar{X}_2 - d_0) / \sqrt{[s_p^2 (1/n_1 + 1/n_2)]}$$

The calculated t statistic is then compared against a Student's t-distribution with **degrees of freedom (df)** defined by:

$$df = n_1 + n_2 - 2$$

2. Dataset Descriptions & Preprocessing

To demonstrate this function, analysis was conducted on two provided datasets containing semi-annual continuous observations from the years **1969 (test1.csv)** and **1970 (test2.csv)**.

Data Context:

- **Dataset 1 (1969):** Comprises two series: *1969 Jan-June* (182 observations) and *1969 July-Dec* (184 observations).
- **Dataset 2 (1970):** Comprises two series: *1970 Jan-June* (181 observations) and *1970 July-Dec* (184 observations).

3. Empirical Results & Analysis

Below are the official outputs structured exactly like the standard spreadsheet analysis function tool (e.g., Excel's *t-Test: Two-Sample Assuming Equal Variances*). Three scenarios are explored to demonstrate different variance structures and outcomes.

Example A: Within-Year Comparison (1969 Jan-June vs. 1969 July-Dec)

This test examines whether the mean values shifted significantly between the first half and the second half of 1969.

Metric	1969 Jan-June	1969 July-Dec
Mean	206.3242	160.9239
Variance	11,266.4855	10,151.9177
Observations	182	184
Pooled Variance		10,706.1396
Hypothesized Mean Difference		0
df		364
t Stat		4.1971
P(T<=t) one-tail		0.000017
t Critical one-tail		1.6491
P(T<=t) two-tail		0.000034
t Critical two-tail		1.9665

Statistical Interpretation (Example A):

Because the absolute value of the *t Stat* (4.1971) is greater than the *t Critical two-tail* (1.9665), and the two-tail p-value ($p < 0.001$) is far below our significance threshold ($\alpha = 0.05$), we **reject the null hypothesis**. There is a statistically highly significant difference between the means of the first and second halves of 1969.

Example B: Across-Year Comparison (1969 Jan-June vs. 1970 Jan-June)

This test examines whether there is a year-over-year change by comparing the first half of 1969 to the first half of 1970.

Metric	1969 Jan-June	1970 Jan-June
Mean	206.3242	181.4365
Variance	11,266.4855	10,444.2473
Observations	182	181
Pooled Variance		10,856.5052
Hypothesized Mean Difference		0
df		361
t Stat		2.2754
P(T<=t) one-tail		0.0117
t Critical one-tail		1.6491
P(T<=t) two-tail		0.0235
t Critical two-tail		1.9666

Statistical Interpretation (Example B):

The **t Stat** of 2.2754 exceeds the two-tail critical value of 1.9666, and the p-value is 0.0235. Since $p < 0.05$, we **reject the null hypothesis** at the 5% significance level. This confirms a significant decrease in the mean value from the first half of 1969 to the first half of 1970.

Example C: Within-Year Comparison (1970 Jan-June vs. 1970 July-Dec)

This test evaluates whether the mean values shifted between the first and second halves of 1970.

Metric	1970 Jan-June	1970 July-Dec
Mean	181.4365	184.5380
Variance	10,444.2473	11,865.5067
Observations	181	184
Pooled Variance		11,160.7500
Hypothesized Mean Difference		0
df		363
t Stat		-0.2804
P(T<=t) one-tail		0.3897
t Critical one-tail		1.6491
P(T<=t) two-tail		0.7793
t Critical two-tail		1.9665

Statistical Interpretation (Example C):

In this scenario, the **t Stat** (-0.2804) is very close to zero, and the two-tail p-value is extremely high (0.7793). Consequently, we **fail to reject the null hypothesis**. There is no statistically significant evidence of a shift in means between the two halves of 1970.

4. Conclusion & Assumptions Validation

The Two-Sample t-Test assuming equal variances requires that the group variances are reasonably similar. Examining our examples:

- **Example A:** Variances are 11,266.49 and 10,151.92 (Ratio: 1.11)
- **Example B:** Variances are 11,266.49 and 10,444.25 (Ratio: 1.08)
- **Example C:** Variances are 10,444.25 and 11,865.51 (Ratio: 1.14)

A general rule of thumb dictates that if the ratio of the larger variance to the smaller variance is less than 2.0, the equal variance assumption holds secure, making this pooled-variance formulation fully reliable and mathematically valid for these datasets.